

Siman Shrestha

Senior Software Engineer, Backend

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Senior Backend Engineer with 5+ years of experience building scalable systems and delivering business impact. Experienced in distributed systems, microservices, observability, and data-driven services. Relocating to NYC.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript

Frameworks: Spring Boot, React, Node.js, Express, NestJS, REST, GraphQL

Databases & Streaming: MySQL, PostgreSQL, MongoDB, Prometheus, Kafka, Redis, Hazelcast, Prisma

Cloud & Tools: AWS, Docker, Kubernetes, Jenkins, Grafana, Github Actions, Splunk, DataDog

AI & Productivity: LangChain, FastMCP, Streamlit, GitHub Copilot, Cline

EXPERIENCE

Visa | Austin, TX

Senior Backend Software Engineer

Sep 2024 – Present

- Built microservices powering Visa Accept, enabling **1M+** merchants across 50 countries to accept tap and QR payments. Designed APIs for merchant onboarding and transaction processing using **Java** and **MySQL**.
- Led the design and implementation of a global transaction fees service for Visa Accept.
- Regularly collaborated with iOS and product teams to define API specifications for client partner needs.
- Enabled comprehensive application monitoring and alerting with **Prometheus** and **Grafana**, driving **99.99%** uptime through automated Active-Active failover and global traffic switching.
- Developed an AI agent using **GPT-4o**, **Python**, and **LangChain** to transform payment data into in-app business insights for merchants, achieving SVP endorsement.
- Built an MCP server with **FastMCP** to expedite bug resolution by aggregating application logs and **Jira** bug reports.

Hinge Health | Austin, TX

Backend Software Engineer II

Aug 2023 – Jul 2024

- Built and scaled microservices at a healthtech startup that manages sensitive user health data for 30M patients, enabling musculoskeletal care delivery at scale using **TypeScript**, **NestJS**, **Postgres**, and **AWS**.
- Accelerated targeted ad campaign delivery by 2.5 days and drove a **6% increase in user registrations** by evolving a hackathon prototype into a production **Kafka** data ingestion pipeline.
- Implemented **envelope encryption** for data at rest in Kafka topics, ensuring **HIPAA and SOC 2 compliance**.
- Built and deployed an asynchronous job queue using **Terraform**, **BullMQ**, and **AWS ElastiCache** that reduced peak service load by **30%** and ensured accurate billing by deduplicating patient records.

Visa | Austin, TX

Senior Fullstack Software Engineer

Oct 2020 – Aug 2023

- Built a real-time metrics dashboard serving **500+** reliability engineers, improving production incident detection time and reducing server load by **5x** using **React**, **Node.js**, **Redis**, and **GraphQL** subscriptions with **WebSockets**.
- Owned and scaled an in-house observability platform used by **1000+** engineers to enable application monitoring, alerting, and logging with **React**, **Node.js**, and **MongoDB**.
- Led deployment of **Grafana** Enterprise in a dual datacenter setup to ensure **99.999%+** availability for critical payment APIs used by **1000+** engineers.
- Implemented a full-stack job queue platform to automate **300+** weekly executions with **Angular**, **Java**, and **MySQL**.

EDUCATION

Texas A&M University, College Station, TX

May 2020

Bachelor of Science in Computer Science, Minor in Visualization